Face Mask Detection using Computer Vision Techniques

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***Abstract*—The use of face masks became essential during global pandemics such as COVID-19. Monitoring mask compliance manually in crowded public environments is inefficient and error-prone. This paper presents a computer vision-based system for detecting whether a person is wearing a face mask or not. The proposed approach combines face detection using Haar Cascade classifiers and classification using a Convolutional Neural Network (CNN). The system processes images by detecting faces, extracting facial regions, and classifying them as masked or unmasked. Experimental results demonstrate that the system can effectively identify mask usage in real-world scenarios with reasonable accuracy.**

***Index Terms*—Computer Vision, Face Detection, Convolutional Neural Network, Mask Detection, OpenCV**

I. Introduction

Face masks became essential during pandemics such as COVID-19 to reduce the spread of infections. Ensuring compliance in public areas such as transportation hubs, offices, and markets is challenging when done manually. Automated systems can significantly improve monitoring efficiency. This project aims to develop a computer vision-based solution to detect whether individuals are wearing face masks in images.

II. Problem Statement

The objective of this work is to detect whether a person is wearing a face mask or not from an input image. Several challenges arise in this task, including variations in face orientation, different types of masks, and varying lighting conditions. These factors make accurate detection difficult and require robust image processing and classification techniques.

III. Proposed Solution

The proposed system consists of three main stages:

1) Face detection using Haar Cascade classifiers
2) Extraction of the detected face region
3) Classification of the face using a Convolutional Neural Network (CNN)

IV. Methodology

A. Dataset

A labeled dataset containing images of individuals with and without face masks was used for training the model.

Fig. 1. Sample images from mask and no-mask dataset

B. Preprocessing

Images were resized to 100×100 pixels and normalized to improve model performance and reduce computational complexity.

C. Model Architecture

A Convolutional Neural Network (CNN) was used for classification. The architecture consists of:

- Convolutional layers for feature extraction
- MaxPooling layers for dimensionality reduction
- Fully connected (Dense) layers for classification

D. Prediction Strategy

A confidence-based thresholding mechanism was used to improve prediction reliability and reduce false positives.

V. Implementation

The system was implemented using the following tools and libraries:

- OpenCV for face detection
- TensorFlow/Keras for model training and inference
- NumPy for data preprocessing and handling

VI. Results

The system was tested on various real-world images. The results indicate that:

- Faces are successfully detected in most cases
- The model correctly classifies mask and no-mask instances
- The system performs reasonably well under normal conditions

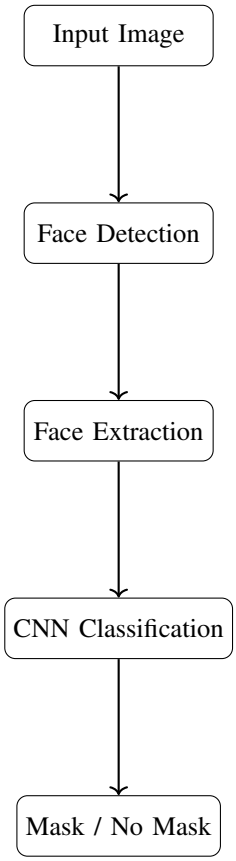

Fig. 2. System Architecture for Face Mask Detection

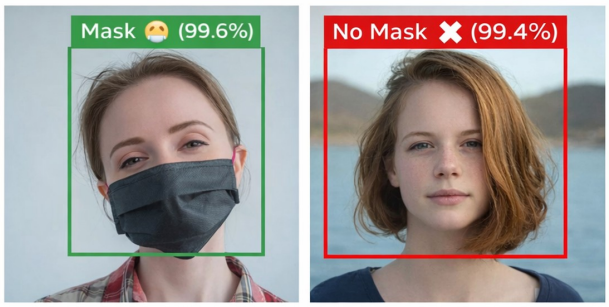

Fig. 3. Mask detection output showing classification results

VII. Challenges Faced

Several challenges were encountered during the development:

- Face detection failures when masks obscure facial features
- Limited dataset diversity affecting model accuracy
- Model bias toward one class during training

VIII. Improvements Made

To enhance system performance, the following improvements were implemented:

- Expansion of bounding boxes to capture full facial regions
- Use of confidence thresholds for more reliable predictions
- Fallback mechanism when face detection fails

IX. Future Work

Future improvements to this system may include:

- Use of deep learning-based face detectors (e.g., MTCNN or YOLO)
- Training on larger and more diverse datasets
- Deployment as a real-time monitoring system using webcam or CCTV

X. Conclusion

This project demonstrates the application of computer vision techniques to solve a real-world problem of mask detection. The system effectively combines face detection and deep learning-based classification. Although simple, the project highlights the importance of dataset quality, preprocessing, and model tuning in achieving reliable results.

References

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